A K-ALGEBRA WITH EXACTLY TWO K-SUBALGEBRAS IS K X K OR K[EPSILON]

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Editorial note. The counted statement is the author's own lemma environment labelled lemma-no-in-between-over-k (source lines 3182-3187); the proof environment begins two lines below it and is its own proof, closing at source line 3199.

Editorial note. The proof uses only elementary algebra available in the retained preamble: no cross-chapter reference and no bibliography entry is invoked inside the retained span, so the package needs neither a source pointer nor a bibliography.

Editorial note. The retained preamble is the chapter's own minimal preamble (sources/stacks-curves/preamble.tex, 217 lines), a verbatim excerpt of the native Xy-pic setup, the native theorem declarations with the mathematical macros, and the project's own no-op definitions of the reference display environments; no macro was invented or redefined.

Editorial note. Section, theorem and equation counters are left at their natural values because the retained text starts at the beginning of the file's preamble, and no initial counter override is needed.

Editorial scope. One original result of the Stacks Project's Algebraic Curves chapter, counted once: a k-algebra over an algebraically closed field with exactly two k-subalgebras is k x k or k[epsilon].

Allowed standard prerequisites (editorial).

- k-subalgebras and intermediate subrings. A k-subalgebra of a k-algebra A is a subring containing k; the hypothesis is that the only such subalgebras are k and A themselves.. Every contradiction in the proof is produced by exhibiting an intermediate subalgebra, at source lines 3192-3193, 3194-3195 and 3195-3196.
- Algebraically closed field and linear factorisation. Over an algebraically closed field k every monic polynomial factors as a product of linear factors t - a_i.. The factorisation of the monic generator P is used at source lines 3194-3197, and the case distinction on the a_i decides the final shape of A.
- Dual numbers k[epsilon]. The k-algebra of dual numbers k[epsilon] with epsilon^2 = 0, as defined in the Varieties chapter and referred to in the surrounding text.. It is one of the two possible conclusions, corresponding to a repeated root of the monic generator (source line 3198).

Lemma 0.1. *Let k be an algebraically closed field. Let $k \subset A$ be a ring extension such that A has exactly two k-sub algebras, then either $A = k \times k$ or $A = k[\epsilon]$.*

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Proof. The assumption means $k \neq A$ and any subring $k \subset C \subset A$ is equal to either k or A. Let $t \in A$, $t \notin k$. Then A is generated by t over k. Hence $A = k[x]/I$ for some ideal I. If $I = (0)$, then we have the subalgebra $k[x^2]$ which is not allowed. Otherwise I is generated by a monic polynomial P. Write $P = \prod_{i=1}^{d}(t - a_i)$. If $d > 2$, then the subalgebra generated by $(t-a_1)(t-a_2)$ gives a contradiction. Thus $d = 2$. If $a_1 \neq a_2$, then $A = k \times k$, if $a_1 = a_2$, then $A = k[\epsilon]$. $\square$